PQFS 2.0 – POST-QUANTUM FILE SHARING SYSTEM

A SECURE, CRYPTOGRAPHICALLY AGILE AND METADATA-PRIVATE FILE SHARING SYSTEM FOR THE QUANTUM ERA

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1. PROBLEM STATEMENT

- Secure file sharing against future quantum attacks and hostile relays.
- Protect confidentiality, integrity and metadata of long-lived secrets.
- Defeat harvest-now-decrypt-later.
- Survive a broken or retired algorithm without re-encrypting.
- Detect tampering, rollback and relay compromise.

2. OBJECTIVES

- Hybrid key protection across three independent math families.
- Threshold t-of-n key sharing over several relays.
- Crypto-agility and a shelf-life policy that picks the tier.
- Threshold custody and verifiable, append-only relay logs.
- Metadata privacy: hide name, exact size and sender.

3. CRYPTOGRAPHIC CONCEPT

Algorithms
- AES-256-GCM chunked encryption
- Hybrid KEM: X25519 + ML-KEM + HQC via HKDF
- Shamir t-of-n over GF(256)
- Ed25519 + ML-DSA-65; SLH-DSA (relays)

Type AEAD + hybrid KEM + threshold sharing + signatures

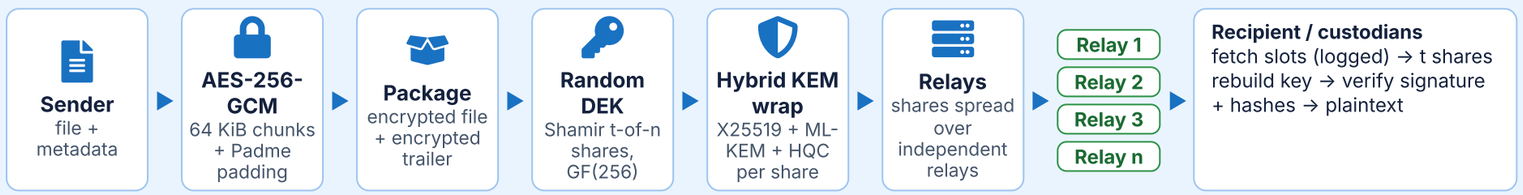
Parameters T1 1-of-1, T2 2-of-3, T3 3-of-5; 64 KiB chunks

Security Confidentiality, integrity, authenticity, rollback resistance, transparency, metadata privacy

4. REAL-WORLD APPLICATIONS

- Long-term document sharing and archiving
- Legal, financial and health records
- Enterprise and key backups
- Board documents with shared custody
- Harvest-now-decrypt-later protection

5. SYSTEM ARCHITECTURE / DESIGN



Each relay keeps an append-only Merkle log (RFC 6962 style) with SLH-DSA signed tree heads; every upload and fetch returns a verifiable receipt.

6. METHODOLOGY / WORKFLOW

1. **Assess** — years + sensitivity pick tier T1/T2/T3
2. **Identities** — hybrid keys; scrypt-protected
3. **Encrypt** — AES-256-GCM chunks + Padme
4. **Wrap + split** — Shamir shares, hybrid KEM, spread
5. **Recover + audit** — t shares, verify, receipts, rewrap

7. TECHNOLOGY STACK

Language	Libraries	Networking	Primitives	Platform	Hardware
Python 3.9+, CLI, about 2,200 lines	cryptography; liboqs-python (Open Quantum Safe)	TCP, JSON messages, PQ-hybrid handshake	AES-GCM, X25519, Ed25519, HKDF, scrypt, SHA3	Command line; tested on Linux	Any PC; relays on localhost or other hosts

PQFS 2.0 | IMPLEMENTATION | ANALYSIS | IMPACT

8. IMPLEMENTATION

- Python CLI **main.py**, 15 commands: keygen, send, receive, approve, rewrap, audit, deprecate, monitor, selftest, demo
- Streaming 64 KiB chunks, constant memory
- Encrypted trailer + signed manifest
- Epoch-based key wraps (crypto-agility)
- 24 built-in self-tests

9. WORKING / OUTPUT (real run, abridged)

```
$ python main.py demo
[POLICY] exposure score 100/100 -> policy tier T3
[SEND] suite ['X25519','ML-KEM-1024','HQC-256']; 3-of-5 shares, 3 relays
[SEND] file 1300055 B padded to 1310720 B (Padme), 20 equal chunks
[RECV] key recovered: epoch 1, tier T3, 3-of-5
[RECV] INTEGRITY: every chunk authenticated, hashes match manifest
  -> byte-for-byte identical to the original
  -> rejected: chunk 11/20 failed authentication (tampered)
  -> refused: topology too weak (3 shares on 1 relay, t=2)
[AUDIT] NEEDS UPGRADE: uses deprecated ['HQC-192']
  -> rejected: ROLLBACK detected (old epoch offered)
[MONITOR] append-only verified: 18 -> 19 entries
DEMO COMPLETE: all checks passed.
```

10. RESULTS & ANALYSIS

24 / 24	12	186 MiB/s	0.1 / 9.6 / 23 ms	0.08 %
unit tests pass	demo scenarios, incl. attacks	encrypt (150 decrypt, verified)	key wrap T1 / T2 / T3	overhead at 50 MiB

#	Test case	Observed result
1	End-to-end round trip (1.3 MB, tier T3)	Byte-for-byte identical to original
2	Ciphertext bit flip	Chunk 11/20 fails authentication; nothing written
3	Weak topology (3 shares, 1 relay, t=2)	Send refused
4	HQC-192 declared broken	Audit flags it; rewrap to epoch 2 (T3), ciphertext untouched
5	Relays hide newer epoch	Rollback detected, file refused
6	Threshold 2-of-3 custody	One custodian alone fails; second approval opens
7	Transparency monitor	Log grew 18 to 19 entries, append-only verified

11. SECURITY ANALYSIS

- **Confidentiality:** all three KEM families must break; fewer than t relays learn nothing.
- **Integrity:** per-chunk AEAD + hashes in signed manifest.
- **Authenticity:** Ed25519 + ML-DSA-65, both verify.
- **Rollback:** epochs, signed minimum tier, MAC.
- **Limits:** not audited; relays see IP/timing; rewrap cannot recall harvested data.

12. CONCLUSION

- Post-quantum security is layered design, not one algorithm swap.
- Three algorithm families, threshold sharing, safe upgrades and logs, at AES speed for bulk data.
- 24 tests and 12 attack/recovery scenarios pass.

13. FUTURE SCOPE

- Independent audit and protocol fuzzing
- Authenticated uploads, rate limits
- Gossip monitors (split-view attacks)
- Multi-host network benchmarks
- Final HQC parameters

14. REFERENCES

- NIST FIPS 203, 204, 205; HQC selection (2025)
- RFC 6962 / 9162 (transparency logs)
- Shamir, How to share a secret (1979)
- Nikitin et al., PURBs / Padme (2019)
- Open Quantum Safe: liboqs
- Code: github.com/theveryholypenguin/pqfs2